Emergence is not automatic. It's a special, precious thing, and provides an enormous simplification when it occurs. Imagine we know the position of every one of the 10^{50} particles in the Earth, but we don't know the momentum of any of them. We possess an enormous amount of information—fully half of the total information available—but we have precisely zero ability to predict where the Earth would be going next. Strictly speaking, even if we know the momentum of all but one of the particles in the Earth, but have no knowledge at all of exactly one momentum, we can't say what the Earth will do next; it's possible that this single particle has as much momentum as all of the others combined.

That's the generic situation in physics. In order to accurately predict what a system made of many parts will do next, you need to keep track of the information of all the parts. Lose just a little bit, and you know nothing. Emergence happens when the opposite is possible: we can throw away almost all the information, keeping just a little bit (as long as you correctly identify which bit), and still say quite a lot about what will happen.

In the case of the center of mass of an object made of many particles, the kind of information in the emergent description we have is exactly the same as the kind we started with (position and momentum), just a lot less of it. But emergence can be more subtle than that; the emergent description may be of an entirely different thing from what we started with.

Consider the air in our room. Imagine that we divide space into tiny boxes, perhaps one millimeter on each side. Each box still contains a huge number of molecules. But instead of keeping track of the state of each one of them, we keep track of average quantities such as the density, pressure, and temperature in each box. It turns out that this is all the information we need to make accurate predictions for how the air will behave. The emergent theory describes a different kind of thing, a fluid rather than a collection of molecules, but that fluid description suffices to describe the air to a high degree of precision. Treating the air as a fluid requires much less data than treating it as a collection of particles; the fluid description is emergent.

Everettian worlds are the same way. We don't need to keep track of the entire wave function to make useful predictions, just what happens in an individual world. To a good approximation we can treat what happens in each world using classical mechanics, with just the occasional quantum intervention when we entangle with microscopic systems in superposition.